Spherical Attention without an MLP: Exact Equilibria, Multistability, and a Stable Internal Cycle

A Reproducible Theoretical and Numerical Report

Anonymous submission

Version of August 16, 2026

Abstract

We study a stack of layers reduced to its attention mechanism: no MLP is present. Each token is a unit-norm point, and each layer moves it and then renormalizes it. We derive a necessary and sufficient condition, valid for any number of tokens, any dimension, and arbitrary score and value matrices: at equilibrium, the transported average seen by each token must be exactly parallel to that token. We give the corresponding finite-layer condition, a reduction by groups, the exact linearization, and, in the tied case $Q^{\mathsf{T}}K = V = V^{\mathsf{T}}$, a complete description in terms of Gram matrices. This description contains consensus, two-sign partitions, multi-vertex shapes, continuous families, and mixtures of eigendirections.

We distinguish finite-layer periodic orbits from periodic orbits of the continuous flow. In the tied symmetric case, a strict Lyapunov function excludes every nonconstant periodic orbit. For independent score and value matrices with nonreciprocal value transport, attention alone admits a stable internal periodic orbit with three tokens. Its return time is 9.12 in normalized depth, and the two transverse Lyapunov exponents are approximately -0.457 and -0.459. All 128 tested perturbations return to the orbit. Across seven dyadic and nondyadic layer strengths, the number of layers per revolution has log–log slope 1.012, as required for convergence to a finite-period continuous orbit. These computations establish existence for one parameter set; they do not determine the frequency of periodic orbits under a random matrix model.

Contents

1 Introduction 3

2 The model, from one layer to continuous time 3

3 Equilibrium equations 4

4 Complete characterization in the tied symmetric case 5

5 Linear stability 7

6 Finite-layer and continuous periodic orbits 8

7 Lyapunov structure of the tied symmetric flow 9

8 A stable periodic orbit produced by attention alone 9

9 Relation to Altafini and Kuehn–Yoon 12

10 Conclusion 13

A Additional proofs 13

B Exact Jacobian on the circle 14

C Unbalanced bipolar split 14

D Reproducibility and claim traceability 15

1 Introduction

A standard Transformer block combines attention, a residual connection, normalization, and an MLP [5]. We study the subsystem obtained by retaining attention, the residual connection, and spherical normalization. The state consists of n unit vectors in $\mathbb{R}^d$; the effective score matrix $S = Q^\mathsf{T}K$ and value matrix V are arbitrary unless symmetry is stated.

Particle descriptions of attention account for cluster formation [2, 3]. Altafini classified four equilibrium classes and derived stability conditions under symmetric V [1]. Kuehn and Yoon studied the tied symmetric model $Q^\mathsf{T}K = V = V^\mathsf{T}$ and obtained eigenspace-selection results [4]. The results below address two complementary questions. First, they characterize every stationary configuration for general S, V, with a closed spectral–Gram formulation in the tied symmetric case. Second, they determine which recurrent motions are compatible with the continuous-depth limit. The equilibrium condition permits nonzero radial attention outputs, a distinction needed to compare multi-vertex equilibria with Altafini's zero-output polygonal class. The tied symmetric flow has a strict Lyapunov function. Untied score and value matrices admit the explicit stable periodic orbit reported in section 8.

2 The model, from one layer to continuous time

2.1 Meaning of the symbols

There are n tokens. Token i is a unit vector $x_i \in \mathbb{R}^d$, hence lies on the sphere $\mathbb{S}^{d-1}$. Two matrices suffice:

- $S = Q^\mathsf{T}K$ determines who attends to whom;
- V determines which vector is actually transported.

The parameter $\beta \geq 0$ controls the sharpness of the choice. At $\beta = 0$, all tokens receive equal weight. As β increases, score differences matter more. For a configuration $X = (x_1, \ldots, x_n)$, define

$$s_{ij}(X) = x_i^\mathsf{T} S x_j, \tag{1}$$

$$a_{ij}(X) = \frac{\exp(\beta s_{ij}(X))}{\sum_{k=1}^n \exp(\beta s_{ik}(X))}, \tag{2}$$

$$y_i(X) = \sum_{j=1}^n a_{ij}(X) V x_j. \tag{3}$$

a_{ij} is the weight assigned by i to j, and y_i is the sum received by i after transport by V.

2.2 A finite layer

Without an MLP, the normalized residual layer is exactly

$$x_i^+ = \frac{x_i + h y_i(X)}{\|x_i + h y_i(X)\|}, \qquad i = 1, \ldots, n, \tag{4}$$

where $h > 0$ is the layer strength. This spherical normalization is an idealized version of RMSNorm [6].

2.3 Many weak layers

Write

$$P_x = I - xx^\mathsf{T}.$$

$P_x z$ retains only the component of z that changes the direction of x; the component parallel to x changes only its length and is removed by normalization. Expanding (4) gives

$$x_i^+ = x_i + hP_{x_i} y_i(X) + O(h^2).$$

When the number of layers grows while h decreases, the limit is

$$\boxed{\dot{x}_i = P_{x_i} y_i(X) = y_i(X) - (x_i^\mathsf{T} y_i(X)) x_i.} \tag{5}$$

The norm remains one because $x_i^\mathsf{T} \dot{x}_i = 0$.

On the circle, write $x_i = (\cos\theta_i, \sin\theta_i)$ and $t_i = (-\sin\theta_i, \cos\theta_i)$. The equation reduces to one scalar per token:

$$\dot{\theta}_i = \sum_{j=1}^{n} a_{ij}(\theta)\, t_i^\mathsf{T} V x_j. \tag{6}$$

All cycle experiments in this report use this equation with three tokens.

3 Equilibrium equations

Theorem 3.1 (Necessary and sufficient condition, general matrices)**.** *A configuration $X \in (\mathbb{S}^{d-1})^n$ is an equilibrium of (5) if and only if, for every token i,*

$$\boxed{y_i(X) = \lambda_i x_i, \qquad \lambda_i = x_i^\mathsf{T} y_i(X).} \tag{7}$$

Equivalently, after removing the softmax denominator,

$$\boxed{\sum_{j=1}^{n} \mathrm{e}^{\beta x_i^\mathsf{T} S x_j} \left[V x_j - (x_i^\mathsf{T} V x_j) x_i \right] = 0, \qquad i = 1, \dots, n.} \tag{8}$$

Proof. The vector field is $P_{x_i} y_i$. It vanishes exactly when y_i lies on the line spanned by x_i. The collinearity coefficient is its inner product with x_i. Finally, the softmax denominator is strictly positive, so multiplying by it or removing it does not change the zero equality. □

The theorem permits $y_i \neq 0$. Any radial output is removed by normalization, so a multi-vertex equilibrium need not have zero attention output.

3.1 Exact condition for a finite layer

Proposition 3.2 (Fixed point of one layer)**.** *Assume $x_i + hy_i \neq 0$. Layer (4) leaves x_i unchanged if and only if*

$$P_{x_i} y_i = 0 \quad \text{and} \quad 1 + h\lambda_i > 0, \qquad \lambda_i = x_i^\mathsf{T} y_i. \tag{9}$$

The first condition is the same as in continuous time. The second is specific to a finite layer: if $1 + h\lambda_i < 0$, normalization sends the point to its antipode $-x_i$; if $1 + h\lambda_i = 0$, it attempts to normalize the zero vector.

3.2 Exact reduction when several tokens coincide

Suppose the configuration contains only q distinct positions $w_1, \dots, w_q$, repeated $r_1, \dots, r_q$ times. Then (8) becomes the finite system

$$\boxed{\sum_{b=1}^{q} r_b e^{\beta w_a^\mathsf{T} S w_b} \left[V w_b - (w_a^\mathsf{T} V w_b) w_a \right] = 0, \quad a = 1, \dots, q.} \tag{10}$$

This is not an approximation: only identical terms have been grouped. For a fixed instance, (10) gives an exhaustive procedure: enumerate the partitions $r_1 + \cdots + r_q = n$, solve the equations, and remove copies related by permutation.

3.3 Polygonal and clustering equilibria

Altafini [1] uses four names. The useful distinction is summarized in table 1.

Table 1: Altafini's four names, expressed using the sum y_i from (3). The last two rows cover all noncollinear shapes.

Name	Configuration	Received-sum condition
Consensus	all x_i equal an eigendirection of V	$y_i \parallel x_i$
Bipolar consensus	each x_i equals u or $-u$, with u an eigenvector	$y_i \parallel x_i$
Polygonal	multiple directions and zero output	$y_i = 0$ for every i
Clustering	every other equilibrium, with or without coincident tokens	$y_i = \lambda_i x_i$, with at least one $\lambda_i \neq 0$

Thus, "the points form a triangle" is a visual description. "Polygonal" in the sense of Altafini's theorem is a much narrower algebraic condition. A triangle whose three outputs are nonzero and radial belongs to his fourth class. Instability of zero-output equilibria says nothing about the stability of that triangle.

4 Complete characterization in the tied symmetric case

We now impose

$$S = V = V^\mathsf{T}. \tag{11}$$

This is exactly the assumption of Kuehn–Yoon [4]. It is stronger than Altafini's: Altafini assumes that V is symmetric with a simple dominant positive eigenvalue, but does not require $Q^\mathsf{T}K$ to be symmetric or equal to V.

Write the tokens as the rows of a matrix $X \in \mathbb{R}^{n \times d}$ and set

$$H = XVX^\mathsf{T}, \qquad R = \exp_\circ(\beta H),$$

where the exponential is applied entrywise. Define

$$M = \mathrm{Diag}(\mathrm{diag}(RH)).$$

Theorem 4.1 (Complete vector equation)**.** *Under* (11),

$$\boxed{X \text{ is an equilibrium} \iff RXV = MX, \qquad \mathrm{diag}(XX^\mathsf{T}) = \mathbf{1}.} \tag{12}$$

This formula is already necessary and sufficient. The next one removes rotations that do not change the problem.

Theorem 4.2 (Spectral description by Gram matrices)**.** *Let $\lambda_1^\star, \dots, \lambda_s^\star$ be the distinct eigenvalues of V, with multiplicities $m_1, \dots, m_s$. Write the component of X in each eigenspace as $C_\alpha \in \mathbb{R}^{n \times m_\alpha}$ and set $G_\alpha = C_\alpha C_\alpha^\mathsf{T}$. Modulo internal rotations within eigenspaces, the equilibria are exactly the families (G_α) satisfying*

$$\boxed{\begin{aligned} &G_\alpha \succeq 0, \qquad \operatorname{rank} G_\alpha \le m_\alpha, \\ &\sum_{\alpha=1}^{s} \operatorname{diag} G_\alpha = \mathbf{1}, \\ &H = \sum_{\alpha=1}^{s} \lambda_\alpha^\star G_\alpha, \qquad R = \exp_\circ(\beta H), \\ &M = \operatorname{Diag}(\operatorname{diag}(RH)), \\ &(\lambda_\alpha^\star R - M) G_\alpha = 0, \qquad \alpha = 1, \dots, s. \end{aligned}} \tag{13}$$

The proof in appendix A only projects (12) onto each eigenspace and factors the matrices G_α. System (13) is a complete characterization, not a preselected list of shapes.

4.1 Families implied by the characterization

Table 2 lists the branches that exist without numerically solving a particular instance.

Table 2: Exact families without an MLP under the tied symmetric assumption.

Situation	Equilibria or families guaranteed immediately
$V = 0$	every configuration in $(\mathbb{S}^{d-1})^n$
One token	all unit spheres contained in eigenspaces of V
Consensus	$x_1 = \cdots = x_n = u$ for every unit eigendirection u
One eigenline	every sign pattern $x_i \in \{u, -u\}$
Kernel of V	every configuration contained in $\ker V$; mixed additions are possible if the active output vanishes
One repeated eigenspace	regular simplices, regular polygons, hypercubes, and other symmetric orbits, with compatible multiplicities
Several eigenspaces	mixed symmetric or asymmetric branches, solved by (13) or (10)

There is therefore no finite list valid for every n, d, V: if $V = 0$, every point is an equilibrium; if an eigenvalue is repeated, continuous families appear. The proper meaning of "all equilibria" is the constructive condition (13).

4.2 Fully explicit classification at $\beta = 0$

At $\beta = 0$, all weights equal $1/n$. Set

$$z = V \sum_{j=1}^{n} x_j.$$

Theorem 4.3 (All equilibria under uniform attention)**.** *For an arbitrary matrix V, a configuration is an equilibrium if and only if one of the following two cases holds:*

1. $z = 0$; every configuration satisfying $V \sum_j x_j = 0$ is admissible;

2. *$z \neq 0$; all tokens equal u or $-u$, their signed sum is nonzero, and u is a real eigenvector of V associated with a nonzero eigenvalue.*

This classification displays both the richness and fragility of the uniform case. With a symmetric matrix, it is an energy dynamics. With a nonsymmetric matrix, a rotation can persist even though the weights never change.

4.3 An exact three-group mixed family

Let $Vu = au$, $Vv = bv$, with $u \perp v$. Take m copies of u, k copies of $qu + \sqrt{1-q^2}v$, and k copies of $qu - \sqrt{1-q^2}v$. This shape is an exact equilibrium precisely when

$$\boxed{q\left[(a-b)\mathrm{e}^{\beta[b+(a-b)q^2]} + (a+b)\mathrm{e}^{\beta[-b+(a+b)q^2]}\right] + a\frac{m}{k}\mathrm{e}^{\beta aq} = 0.} \tag{14}$$

This single equation produces branches that use two eigendirections simultaneously. It explains why "choose one eigenvalue" is insufficient to describe every destination.

5 Linear stability

Condition (8) tells us where the dynamics can stop. Stability tells us what happens after a small positional error.

Let δx_i be a small perturbation perpendicular to x_i. Define

$$\delta s_{ij} = \delta x_i^\mathsf{T} S x_j + x_i^\mathsf{T} S \delta x_j, \tag{15}$$

$$\delta a_{ij} = \beta a_{ij}\left(\delta s_{ij} - \sum_k a_{ik}\delta s_{ik}\right), \tag{16}$$

$$\delta y_i = \sum_j a_{ij} V \delta x_j + \sum_j \delta a_{ij} V x_j. \tag{17}$$

At an equilibrium $y_i = \lambda_i x_i$, the tangent variation of the vector field is

$$\boxed{\delta \dot{x}_i = P_{x_i}\delta y_i - \lambda_i \delta x_i.} \tag{18}$$

These equations give the exact stability matrix for arbitrary S, V. The equilibrium is locally attracting if every tangent eigenvalue has strictly negative real part. A zero eigenvalue may merely correspond to a continuous family generated by rotation; it must then be removed before drawing a conclusion.

5.1 Pure eigendirections in the tied symmetric case

For consensus $x_i = u$ on an eigenvalue λ_p, the local condition is especially simple:

$$\lambda_p > 0, \qquad \lambda_k < \lambda_p \quad (k \neq p). \tag{19}$$

Thus stable consensus uses the largest simple positive eigenvalue.

For a balanced split between u and $-u$, the thresholds are

$$\lambda_p > 0: \qquad \lambda_k < \lambda_p \tanh(\beta\lambda_p) \qquad \text{for every } k \neq p, \tag{20}$$

$$\lambda_p < 0: \qquad \lambda_p < \lambda_k < \lambda_p \tanh(\beta\lambda_p) \qquad \text{for every } k \neq p. \tag{21}$$

The second case necessarily selects the most negative eigenvalue. Formulas for an unbalanced split are given in appendix C.

5.2 A stable nonzero-output triangle

For $V = S = \mathrm{Diag}(2, -3)$, $\beta = 1.5$, and three tokens, root finding gives the angle configuration, rounded here,

$$(0,\ 1.546889,\ 4.736296).$$

Its least-contracting linear rate is -0.9144, and the residual computed before rounding is 1.3×10^{-15}. More importantly, the norms of the three unnormalized outputs are approximately 40.27, 269.79, and 269.79: they are nonzero and point radially. This stable triangle therefore belongs to Altafini's "clustering" class, not his "polygonal" class.

The multistart small-system search gives figure 1. Two independent seeds recover exactly the same sets after removing permutations and global sign flips.

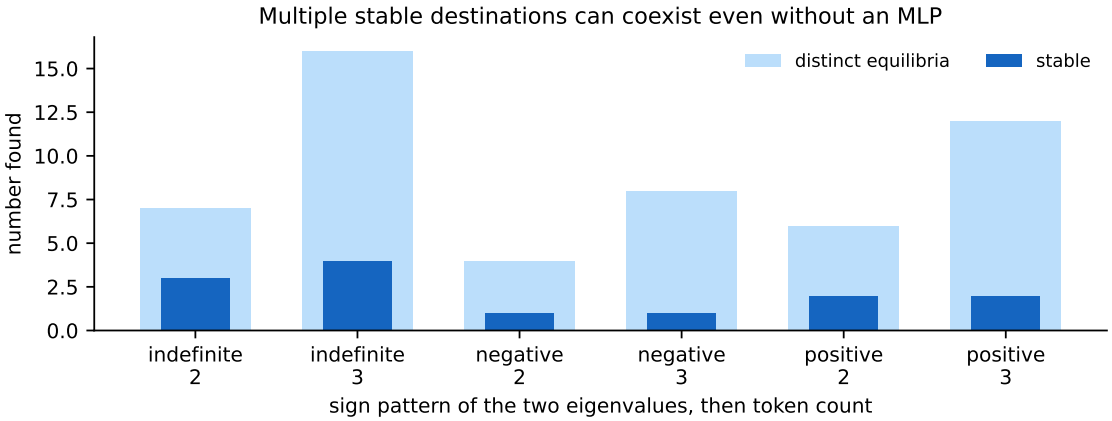


Figure 1: Catalogue without an MLP for symmetric $V = S$ in dimension two. The spectra are $(2, -3)$ with $\beta = 1.5$, $(-0.4, -4)$ with $\beta = 0.03$, and $(3, 2)$ with $\beta = 1.5$. The second line of each label is the token count. Numerical search discovers the branches; theoretical completeness comes from (13), not from the number of starts.

6 Finite-layer and continuous periodic orbits

6.1 Finite-layer periodic orbits

A period-p cycle of the layer is a sequence of p distinct configurations that returns exactly to its starting point after p applications of (4). This object depends on the layer strength h.

Lemma 6.1 (A fixed number of layers does not survive unchanged)**.** *If the layer satisfies $F_h(X) = X + hf(X) + O(h^2)$ and p remains fixed as $h \to 0$, then every limit of a period-p cycle satisfies $f(X) = 0$.*

Indeed, $F_h^p(X) = X + phf(X) + O(h^2)$. A three-layer cycle can therefore collapse to an equilibrium as layers are weakened. It does not prove the existence of a continuous-time loop.

6.2 Continuous periodic orbits

A continuous cycle is a loop traversed in a time $T > 0$. A layer of strength h covers only a fraction of it, so the number of layers per revolution must grow as T/h. This is the signature measured in figure 4.

6.3 Rigid rotations and internal motion

If all inner products $x_i^\mathsf{T} x_j$ remain constant while the entire set rotates, the shape does not evolve: this is a rigid rotation. In an internal cycle, at least one inter-token distance changes and then returns. The latter is a stronger phenomenon.

An analytical example already shows that attention without an MLP can rotate. Take one token on the circle, $S = 0$, and

$$V = \omega \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Then $\dot{x} = Vx$, and the token rotates with period $2\pi/|\omega|$. This rotation is neutral: it does not show that a loop attracts perturbations. The experiment in section 8 establishes precisely this harder point.

7 Lyapunov structure of the tied symmetric flow

Under $S = V = V^\mathsf{T}$ and for $\beta > 0$, define

$$E_\beta(X) = \frac{1}{2\beta} \sum_{i,j=1}^{n} \exp\left(\beta x_i^\mathsf{T} V x_j\right). \tag{22}$$

If $Z_i = \sum_j \exp(\beta x_i^\mathsf{T} V x_j)$, direct calculation gives

$$\frac{dE_\beta}{dt} = \sum_{i=1}^{n} Z_i \, \|\dot{x}_i\|^2 \geq 0. \tag{23}$$

The quantity increases strictly whenever a token moves. A closed trajectory would have to return to its initial value of E_β, which is impossible unless it is already stationary. This is the weighted gradient structure used by Kuehn–Yoon [4].

At $\beta = 0$, the same conclusion holds with

$$\Phi(X) = \frac{1}{2n} \left(\sum_i x_i\right)^\mathsf{T} V \left(\sum_j x_j\right).$$

The identity uses the full relation $S = V = V^\mathsf{T}$, rather than symmetry of V alone. For independent S and V, the calculation does not produce a Lyapunov function.

On the circle at $\beta = 0$, the departure from reciprocity is seen exactly in

$$\frac{\partial \dot{\theta}_i}{\partial \theta_j} - \frac{\partial \dot{\theta}_j}{\partial \theta_i} = \frac{1}{n} t_i^\mathsf{T} (V - V^\mathsf{T}) t_j. \tag{24}$$

The antisymmetric part of V therefore supplies circulation. At $\beta > 0$, row-wise normalization of the weights adds further terms. Even when ordinary cross-derivatives do not match, the case $S = V = V^\mathsf{T}$ remains cycle-free because of the weighted metric in (23). This detail prevents conflating a Euclidean gradient with a monotone quantity.

8 A stable periodic orbit produced by attention alone

8.1 Explicit parameters

The model uses three tokens on the circle, no MLP, and

$$S = \begin{pmatrix} 1.112495 & 1.768642 \\ -2.540753 & 3.294286 \end{pmatrix}, \qquad V = \begin{pmatrix} -1.809502 & 1.419182 \\ -1.827949 & -3.611608 \end{pmatrix}, \tag{25}$$

with $\beta = 1.110063$. The audit measures depth with a scale $\rho = 0.551488$: its equation is $\dot{\theta}_i = \rho \sum_j a_{ij} t_i^{\mathsf{T}} V x_j$. Absorbing ρ into time gives exactly (6).

This model was isolated by ablating a candidate from a broader search with an MLP. After all MLP parameters are removed, the cycle persists. Conversely, the MLP alone, with attention removed, converges to a fixed point. This model-selection step matters: it establishes existence of an attention cycle, not its frequency under a random matrix law.

8.2 Internal geometry along the orbit

Independent fourth-order Runge–Kutta integration with step 0.005 gives return time $T = 9.12$ in normalized depth, or 5.030 in the time of (6) after absorbing ρ. The 90th-percentile return residual is 1.56×10^{-3} radians.

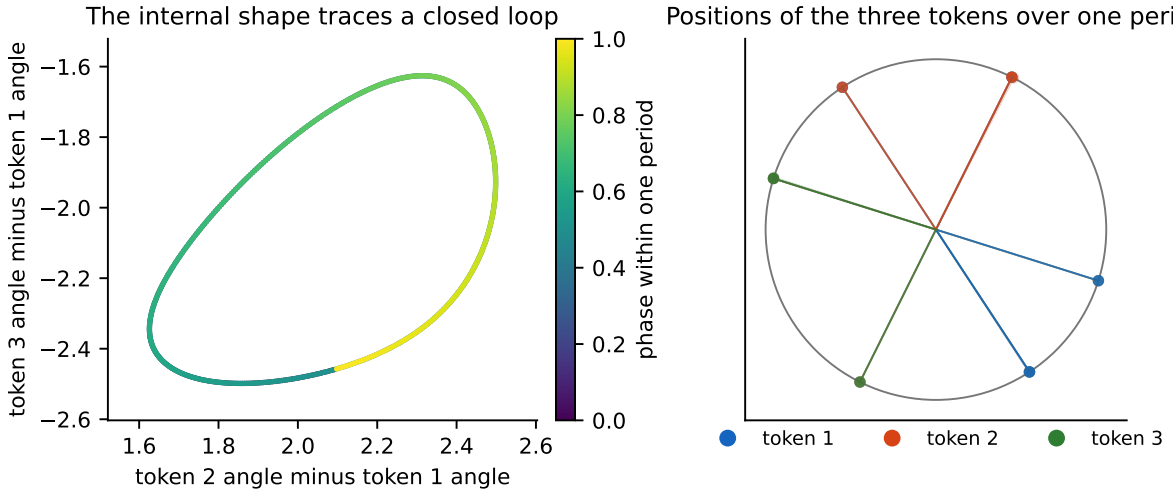


Figure 2: Internal cycle produced by attention alone. Left: the two relative angles trace a closed loop; a rigid rotation would be a point. Right: seven phases of one revolution show the three tokens on the circle. Angles are always taken modulo 2π.

All three pairwise inner products vary from -0.8002 to -0.0551. The shape therefore changes substantially. The weights sent by token 1 to tokens 1, 2, and 3 respectively span

$$[0.298, 0.982], \qquad [0.015, 0.701], \qquad [0.00018, 0.0341].$$

8.3 Transverse stability

On an autonomous closed orbit, a perturbation that merely advances the phase is neither contracted nor amplified: its theoretical rate is zero. The spectrum measured over 2000 time units is

$$(-0.00273,\ -0.45676,\ -0.45937). \tag{26}$$

The first number is the finite approximation to the phase zero. The other two are clearly negative: deviations transverse to the orbit contract.

We also added 128 Gaussian perturbations with standard deviation 0.08 around one point of the cycle. The median distance to the entire orbit decreases from 0.0525 to 0.00194; its 90th percentile decreases from 0.0938 to 0.00320. The terminal plateau is of the same order as the spacing used to store the reference orbit.

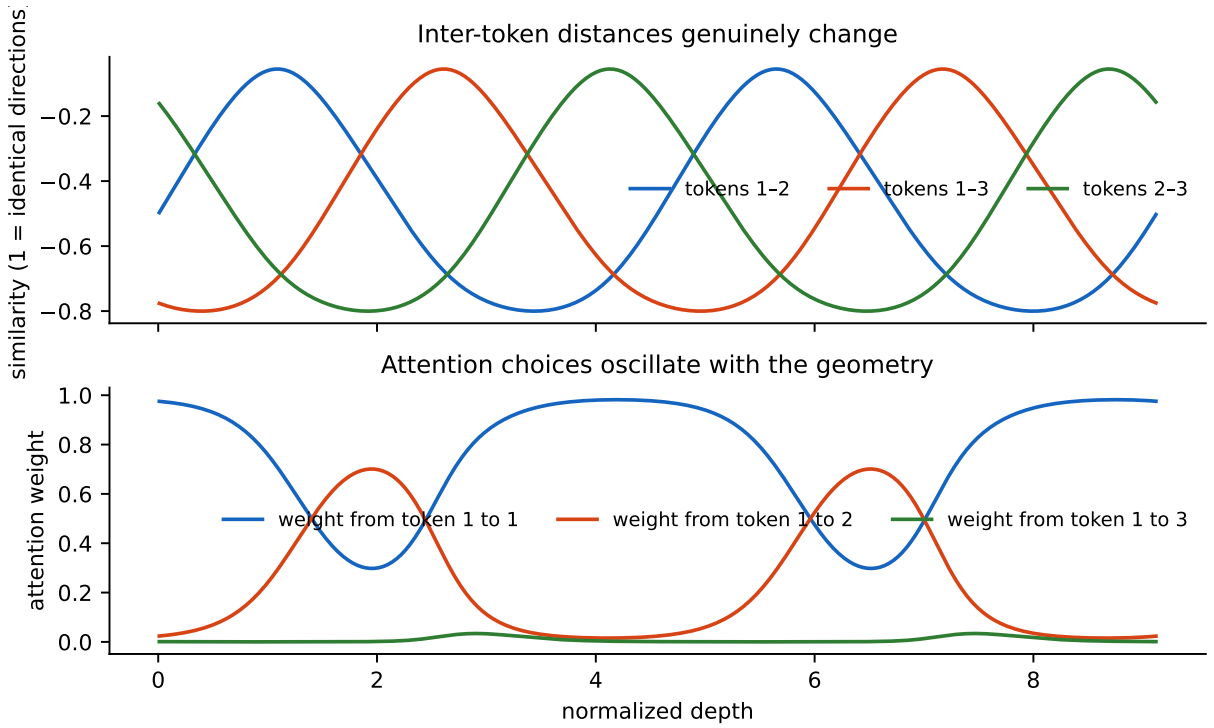


Figure 3: One revolution of the cycle. Top: pairwise token inner products oscillate. Bottom: the weights in the first attention row oscillate as well. The second period visible in the distances comes from a role permutation among tokens during the full return of the labeled angles.

8.4 Scaling of the discrete period with layer strength

We replayed the exact layer (4) at seven ratios, including three nondyadic values. Table 3 reports the best complete return.

Table 3: Return of the attention-only cycle as each layer is weakened. The time per revolution is the number of layers multiplied by the ratio.

Ratio	Layers per revolution	Revolution time	Maximum angular error
1/16	138	8.625	2.38×10^{-2}
1/31	275	8.871	9.46×10^{-3}
1/64	576	9.000	5.38×10^{-3}
1/127	1151	9.063	5.43×10^{-5}
1/256	2328	9.094	1.13×10^{-3}
1/509	4635	9.106	7.42×10^{-4}
1/1024	9333	9.114	1.09×10^{-5}

The slope of log(layers per revolution) against log(1/ratio) is 1.012. The revolution time approaches the continuous equation's period while the layer count diverges.

8.5 Ablations

Table 4 prevents an overly broad conclusion. The three most-studied chaotic candidates do not remain chaotic after the MLP is removed; they converge to an equilibrium in the attention-only ablation.

These tests establish the existence of a stable internal cycle produced by attention alone. They

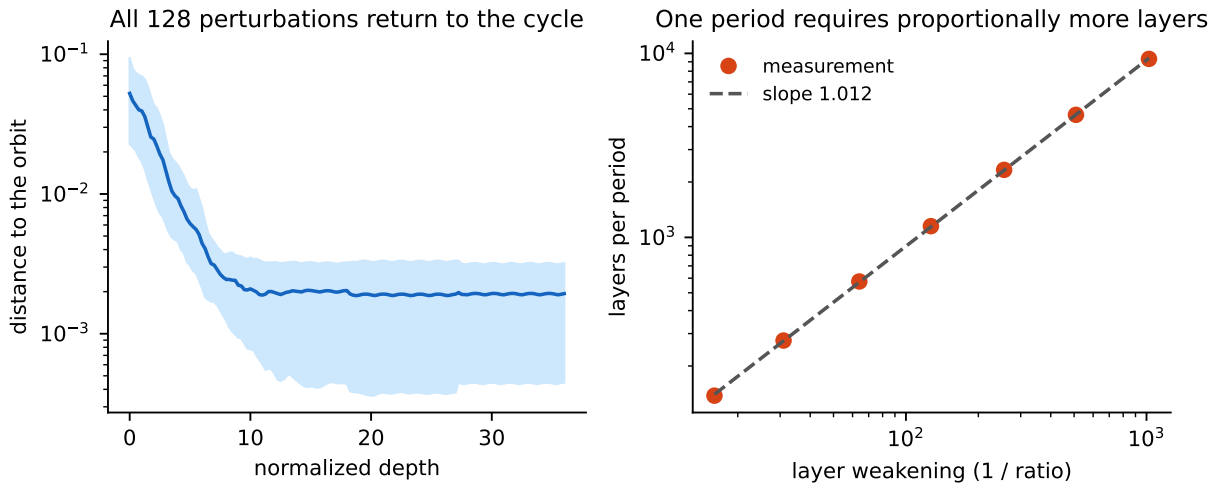


Figure 4: Left: contraction of 128 perturbations toward the orbit. The line is the median and the band contains 80% of trials. Right: number of layers per revolution; the near-unit slope is the signature of a continuous cycle rather than a three- or four-layer artifact.

Table 4: Attention-only ablation results for the representatives studied in detail.

Origin of representative	Attention structure	Result without an MLP
Three-token cycle, index 1323	general S, general V, $\beta > 0$	stable internal cycle
Strong three-token chaos, index 2612	symmetric S, general V, $\beta > 0$	fixed consensus
Weak three-token chaos, index 1051	$S = V = V^{\mathsf{T}}$, $\beta > 0$	fixed consensus
Four-token intermittency, index 3118	symmetric S, general V, $\beta = 0$	fixed equilibrium

do not estimate its frequency under random S, V. None of the targeted ablations produced an attention-only chaotic attractor.

9 Relation to Altafini and Kuehn–Yoon

9.1 Altafini's assumptions

Altafini's model does contain the matrices Q and K. His main analysis assumes

$$V = V^{\mathsf{T}}, \qquad \lambda_1 > 0 \text{ simple and strictly dominant.} \tag{27}$$

It does not assume that $Q^{\mathsf{T}}K$ is symmetric. Under these conditions, Altafini proves that the zero-output equilibria he calls polygonal are unstable. He also shows that bipolar consensus and clustering equilibria can be stable and coexist.

9.2 Compatibility with polygonal instability

There are three independent reasons:

1. the stable triangle in section 5 has nonzero outputs. It belongs to Altafini's "clustering" class, which his theorem does not declare unstable;
2. the negative-definite branches of Kuehn–Yoon have no positive eigenvalue and therefore fall outside (27);
3. the cycle in (25) uses nonsymmetric V and is not an equilibrium. It lies outside both the assumption on V and the question of "stability of a polygonal equilibrium."

9.3 Consequences of the Kuehn–Yoon symmetry

Kuehn–Yoon impose the stronger relation

$$Q^{\mathsf{T}}K = V = V^{\mathsf{T}}.$$

It supplies the monotone quantity (22). Their paper proves global selection results inside a cone when the dominant positive eigenvalue exceeds all others in magnitude, and for two tokens when V is negative definite [4]. Our characterization (13) addresses the different question "what are all possible stationary points?", including mixed branches and multiplicities, without claiming that every one is reached from a generic initialization.

Table 5: Scope of the three analyses.

	Altafini	Kuehn–Yoon	This report
MLP	absent	absent	absent
Score $S = Q^{\mathsf{T}}K$	general	equal to V	general, then the case equal to V
Value V	symmetric, simple positive top eigenvalue	symmetric, selected spectra	general; complete classification when tied symmetric
Equilibria	four classes, thresholds on pure branches	spectral selection and pure-branch stability	necessary and sufficient condition, group reduction, spectral–Gram system
Cycles	possible cycle noted when V loses symmetry	excluded by the energy	explicit stable internal cycle without an MLP

10 Conclusion

Equation (8) characterizes every equilibrium for arbitrary n, d, S, V. Under $S = V = V^{\mathsf{T}}$, (13) gives an equivalent spectral–Gram system and (23) excludes nonconstant recurrent orbits. These statements include stable multi-vertex configurations with nonzero radial output and are consistent with Altafini's instability theorem for the narrower zero-output polygonal class.

For untied S, V, the matrices in (25) yield a stable internal periodic orbit. Its transverse spectrum, perturbation experiment, and discrete-period scaling establish persistence in the continuous-depth limit. The computation is an existence result for one parameter set. A probabilistic theory for periodic or chaotic attention-only dynamics under random matrices remains open.

A Additional proofs

A.1 Proof of the spectral–Gram theorem

In an eigenbasis of V, write $X = C_1 \oplus \cdots \oplus C_s$. Projecting $RXV = MX$ onto the eigenspace of $\lambda_\alpha^\star$ gives

$$(\lambda_\alpha^\star R - M)C_\alpha = 0.$$

Multiplication on the right by C_α^{T} yields $(\lambda_\alpha^\star R - M)G_\alpha = 0$. The identities for H and the diagonals follow from the orthogonal decomposition.

Conversely, factor any positive solution as $G_\alpha = C_\alpha C_\alpha^\mathsf{T}$ with at most m_α columns. The image of G_α is the same as that of C_α, so the equation for G_α implies the equation for C_α. Placing these factors in orthogonal eigenspaces reconstructs X and $RXV = MX$. Two factorizations of the same G_α differ by an internal rotation of the eigenspace, which proves the statement modulo those rotations.

A.2 Proof of the classification at $\beta = 0$

Under uniform attention, every vector field has the form $P_{x_i} z/n$, where $z = V \sum_j x_j$. If $z = 0$, the configuration is stationary. Otherwise every x_i must be parallel to z, hence $x_i \in \{u, -u\}$ with $u = z/\|z\|$. The sum equals cu. Then $z = cVu$ is parallel to u, so u is an eigenvector. Since $z \neq 0$, both c and the corresponding eigenvalue are nonzero. The converse is immediate.

A.3 No zero outputs when V is positive definite

Under $S = V \succ 0$, group the q distinct positions w_a. The matrix

$$\mathcal{R}_{ab} = \exp\Big(\beta \left\langle V^{1/2} w_a, V^{1/2} w_b \right\rangle\Big)$$

is strictly positive definite for distinct points and therefore invertible. The zero-output condition is $\mathcal{R}\,\mathrm{Diag}(r_1, \ldots, r_q) W V = 0$. Since $\mathcal{R}$, $\mathrm{Diag}(r_1, \ldots, r_q)$, and V are invertible, this equality would force $W = 0$, contradicting the unit norm of its rows. The multi-vertex equilibria that remain possible must therefore have nonzero radial output.

B Exact Jacobian on the circle

This appendix gives a directly programmable formula. Set

$$g_{ij} = t_i^\mathsf{T} V x_j, \qquad s_{ij} = x_i^\mathsf{T} S x_j, \qquad f_i = \sum_j a_{ij} g_{ij}.$$

For angle θ_k,

$$D_k s_{ij} = \mathbf{1}_{i=k}\, t_i^\mathsf{T} S x_j + \mathbf{1}_{j=k}\, x_i^\mathsf{T} S t_j, \tag{28}$$

$$D_k g_{ij} = -\mathbf{1}_{i=k}\, x_i^\mathsf{T} V x_j + \mathbf{1}_{j=k}\, t_i^\mathsf{T} V t_j. \tag{29}$$

Differentiating the softmax then gives

$$\boxed{J_{ik} = \sum_j a_{ij} \left[D_k g_{ij} + \beta g_{ij} \left(D_k s_{ij} - \sum_\ell a_{i\ell} D_k s_{i\ell} \right) \right].} \tag{30}$$

At an equilibrium, the real parts of the eigenvalues of J determine stability. For a finite layer, one directly computes the Jacobian of F_h; its eigenvalues must have modulus below one. For small h, a continuous rate λ gives a layer multiplier $1 + h\lambda + O(h^2)$.

C Unbalanced bipolar split

Let n_+ be the number of $+u$ tokens, n_- the number of $-u$ tokens, and $r = n_+/n_-$. For an eigenvalue λ_p, define

$$c = \mathrm{e}^{2\beta\lambda_p}, \qquad \sigma(c, r) = \frac{(c - r)(cr - 1)}{r(c^2 - 1)}.$$

If $\lambda_p > 0$, the split is locally stable exactly when

$$\lambda_p > \frac{|\log r|}{2\beta}, \qquad \lambda_k < \lambda_p \sigma(c, r) \quad (k \neq p).$$

If $\lambda_p < 0$, the condition is

$$\lambda_p < -\frac{|\log r|}{2\beta}, \qquad \lambda_p < \lambda_k < \lambda_p \sigma(c, r) \quad (k \neq p).$$

For $r = 1$, $\sigma = \tanh(\beta\lambda_p)$ and we recover (20)–(21).

D Reproducibility and claim traceability

The main files are:

- `experiments/spectral_self_attention/non_mlp_report_experiments_en.py` for reintegration, perturbations, the depth test, and the English figures;
- `data/spectral_self_attention/non_mlp_report_results.json` for every number displayed in the cycle section;
- `data/spectral_self_attention/continuous_ode_f4_p3_cycle_attention_only.json` for the initial ablation;
- `data/spectral_self_attention/continuous_lyapunov_spectrum_f4_p3_cycle_attention_only_relaxed2.json` for the long spectrum;
- `data/spectral_self_attention/equilibria/planar_small_systems.csv` and `search_saturation.csv` for the small-system catalogue;
- `experiments/spectral_self_attention/equilibrium_catalogue.py` and `run_equilibrium_catalogue.py` for equilibrium roots.

From the repository root, the command

```
uv run python \
  experiments/spectral_self_attention/non_mlp_report_experiments_en.py
```

recreates the four English figures and the JSON summary. The matrices in (25), random seeds, step sizes, durations, and thresholds are stored in these files.

Table 6: Exact status of the main claims.

Claim	Status
Condition (8)	necessary and sufficient algebraic theorem
System (13)	theorem, complete modulo eigenspace rotations
No cycle when $S = V = V^\mathsf{T}$	exact consequence of (23)
Displayed stable triangle	numerical root, residual $< 1.3 \times 10^{-15}$, contracting Jacobian
General cycle without an MLP	numerical evidence: return, spectrum, perturbations, and depth law
Absence of chaos without an MLP	not established; only observed in the targeted ablations
Frequency of random cycles	not measured in a dedicated survey

References

[1] Claudio Altafini. Multistability of self-attention dynamics in transformers. *arXiv preprint arXiv:2511.11553*, 2025.

[2] Borjan Geshkovski, Clément Letrouit, Yury Polyanskiy, and Philippe Rigollet. The emergence of clusters in self-attention dynamics. In *Advances in Neural Information Processing Systems*, volume 36, pages 57026–57037, 2023.

[3] Borjan Geshkovski, Clément Letrouit, Yury Polyanskiy, and Philippe Rigollet. A mathematical perspective on transformers. *Bulletin of the American Mathematical Society*, 62(3):427–479, 2025.

[4] Christian Kuehn and Jaeyoung Yoon. Spectral selection in symmetric self-attention dynamics. *arXiv preprint arXiv:2604.26085*, 2026.

[5] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[6] Biao Zhang and Rico Sennrich. Root mean square layer normalization. In *Advances in Neural Information Processing Systems*, volume 32, 2019.